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324555(24)

B. E. (Fifth Semester) Examination, April-May 2016

(New Scheme)

(Elect. Branch)

CONTROL SYSTEM ENGINEERING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all questions. Part (a) of each question is compulsory. Attempt any two from (b), (c) & (d) of each question.*

Unit-I

(a) What is Feedback? Explain one effect of feedback. 2

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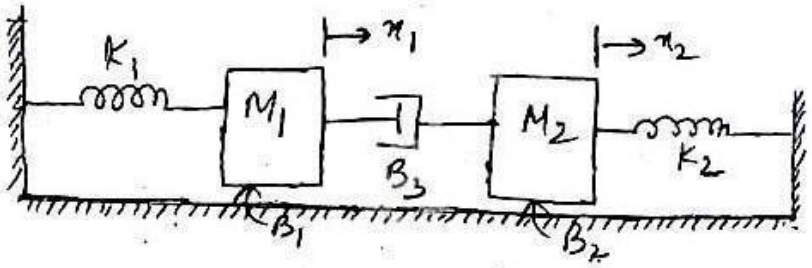
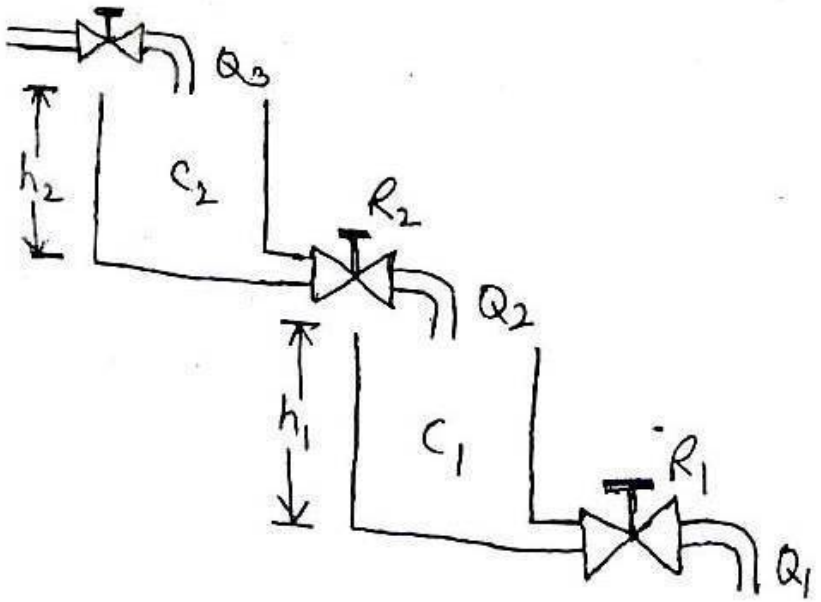
(b) Write the performance equation for the system shown and find the transfer function $X_2(s)/F(s)$. 7



(c) The process plant consists of two tanks of capacitances C_1 and C_2. If the flow rate 7



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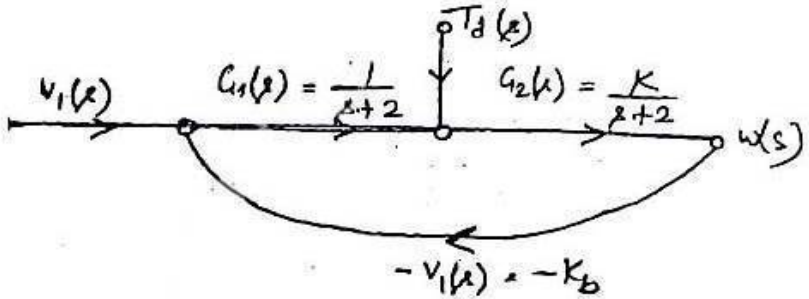
into the top tank is Q_3, find the transfer function relating this flow to the level is the bottom tank. Each tank has a resistance R is the outlet pipe. Consider the tanks to be non-interacting.

(d) The SFG of the figure shown below describes a speed control system using an armature controlled d.c. motor. The output of the system is speed $w(n)$ while the input is voltage $V_1(s)$. The disturbance $T_d(n)$ is present in the forward path of the system. Determine the sensitivity of the system $S_K^{M_s}$ and $S_{K_b}^{M_s}$. 7



Unit-II

(a) What is a servomotor? 2

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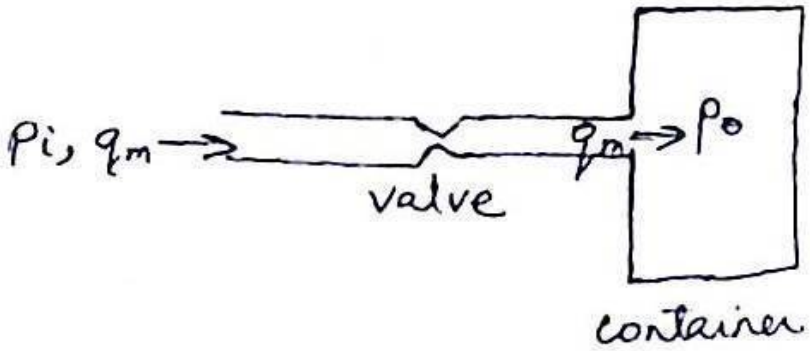
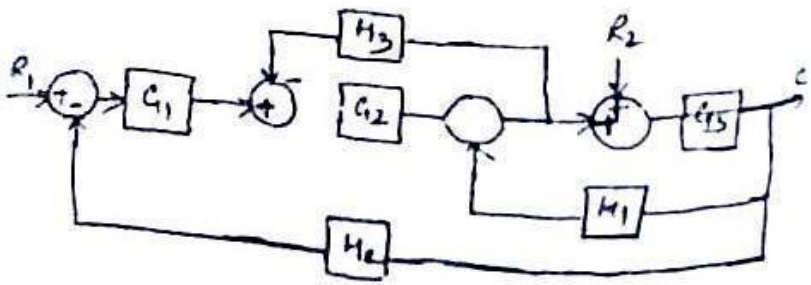
(b) Develop a mathematical model the pressure change P_0 is a rigid contained as a function of the inlet pressure change P_i when air passes through a valve into the container, assuming the process to be isothermal where P_0 and P_i are small variation from the steady stele pressure P_S. 7



(c) Obtain the output of the system shown. 7



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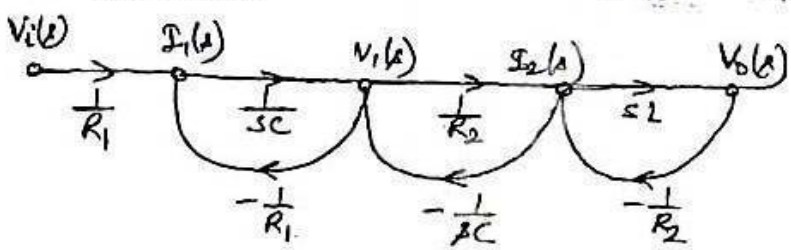
(d) Obtain the transfer function $\frac{V_0(s)}{V_i(s)}$ using Mason's gain formula. 7



Unit-III

3. (a) What is meant by characteristics equation? 2

(b) For a unity feedback system having open loop transfer function as :

$$G(s)=\frac{K(s+2)}{s^2\left(s^2+7s+12\right)}$$

Determine :

(i) Type an order of the system. 1

(ii) Error constants 3

(iii) Steedy state error for parabolic input 3

(c) The closed loop transfer function of a system is given by :

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$$\frac{C(s)}{R(s)}=\frac{K}{s\left(s^2+s+1\right)(s+2)+K}$$

Determine the value of K for which the system is stable. 7

(d) For a unity feedback system, the open loop transfer function is given by :

$$G(s)=\frac{K(s+2)(s+3)}{s(s+1)}$$

Sketch the root loci of the system when K is varied from 0 to ∞. 7

Unit-IV

4. (a) State the Nyquist stability criterion. 2

(b) Obtain the polar plot. Obtain the frequency and magnitude $|G(jw)|$ at which the polar plot intersects the real and imaginary axis. The system given is :

$$G(s)=\frac{K}{(1+sT_1)(1+sT_2)}$$

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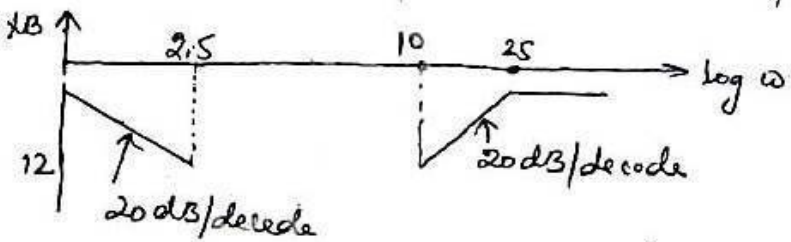
(c) Determine the gain cross-over frequency phase cross over frequency, gain margin and phase margin of a system will open loop transfter function

$$G(s)=\frac{1}{s(1+s)(1+2s)}$$

Comment on the stability of the system. (Without using Bode). 7

(d) Find the transfer fucntion of a system whose Bode plot is as : 7



Unit-V

(a) What is Compensation? 2

(b) Consider a lag-lead comparator $G_C(s)$ defined by

$$G_C(s)=K_C\frac{\left(s+\frac{1}{T_1}\right)\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{\beta}{T_1}\right)\left(s+\frac{1}{\beta T_2}\right)}$$

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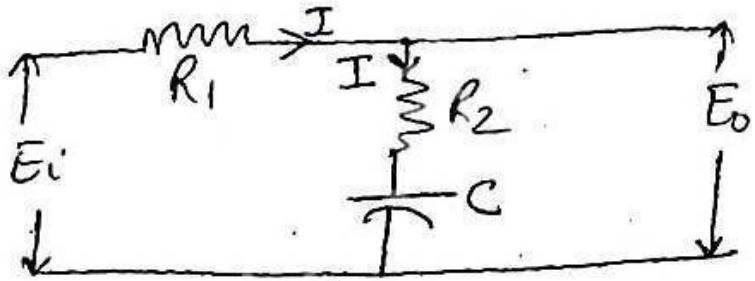
Show that at a frequency w_1 where $w_1=\frac{1}{\sqrt{T_1T_2}}$, the phase angle $G_C(jw)$ becomes zero. 7

(c) For a phase-leg compensator shown



Show that w_m is the geometric mean of the two corner frequencies of the compensator where w_m is the frequency where the maximum phase-lag angle ϕ_m occurs. 7

(d) Explain the effects of : 7

(i) Phase lag compensation

(ii) Phase lead compensation

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